Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

CHEMISTRY

Class – 7

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1. Changes Around Us

Solutions

Worksheet – 1

1. **Correct answer – B) Slow change**
   Explanation: Growth of a plant takes days, weeks, or even months. Such changes occur gradually over time, so they are slow changes.
2. **Correct answer – C) Rubber band stretching — Reversible change**
   Explanation: A reversible change is one that can be undone. Stretching a rubber band is reversible because it returns to its original shape when released.
3. **Correct answer – B) Desirable**
   Explanation: Desirable changes are useful or beneficial. Drying clothes helps make them ready to wear and is useful.
4. **Correct answer – C) Irreversible**
   Explanation: Once clay is baked, it becomes hard and cannot revert to soft clay. This is an irreversible change.
5. **Correct answer – C) Germination of seeds**
   Explanation: Desirable changes are beneficial or useful. Germination is useful as it produces plants.
6. **Correct answer – C) Undesirable**
   Explanation: Undesirable changes are harmful or unwanted. Milk spoilage produces bad odor and is unsafe for consumption.
7. **Correct answer – B) Periodic change**
   Explanation: Changes that occur repeatedly at fixed intervals are periodic changes.
   Example: Day and night cycle.
8. **Correct answer – C) Day and night**
   Explanation: Day and night occur regularly every 24 hours → periodic change.
9. **Correct answer – D) Sneezing**

   Explanation: Non-periodic changes do not follow a regular time pattern. Sneezing occurs randomly.
10. **Correct answer – C) Stretching a rubber band**
    Explanation: Reversible changes can return to their original form without producing a new substance. Stretching a rubber band does not produce a new material.

Worksheet-2

1. **Correct answer –B) Evaporation occurs at surface, boiling occurs throughout the liquid**

   Explanation:
   - Evaporation happens only at the surface of a liquid and can occur at any temperature.
   - Boiling occurs throughout the liquid and happens only at the boiling point.
     Example: Water in a pond evaporates slowly; water in a pan boils at 100°C.
2. **Correct answer –A) Physical and reversible**

   Explanation: Magnetizing iron does not change its chemical composition; the magnetism can be reversed by demagnetizing.
3. **Correct answer – B) Chemical change**

   Explanation: Photosynthesis produces glucose and oxygen from carbon dioxide and water, forming new substances.
4. **Correct answer – C) Shattering of a mirror**

   Explanation: Although it's a physical change (no new substance is formed), it cannot easily be reversed.
   - Freezing and melting of water, and boiling water, are reversible physical changes.

5. **Correct answer – C) Atoms are rearranged to form new substances**
   Explanation: Chemical changes involve rearrangement of atoms to produce new substances with different properties.
6. **Correct answer – B) Because the chemical composition of the glass remains unchanged**
   Explanation: Breaking a mirror changes shape, not the chemical composition..
7. **Correct answer – A) Evaporation**
   Explanation: Evaporation occurs gradually at the surface and below the boiling point.
8. **Correct answer – A) Mass**
   Explanation: Physical changes may change state, size, or shape, but mass remains constant.
9. **Correct answer –B) It decomposes into ammonia and hydrogen chloride**
   Explanation: Ammonium chloride sublimes, breaking into NH_3 and HCl gases, which is a chemical change.
   - Physical change: solid → gas (sublimation)
10. **Correct answer – D) Because it converts carbon dioxide and water into oxygen and glucose**
    Explanation: Photosynthesis produces glucose (food) and oxygen for all living organisms, which are new substances, making it a chemical change.

Competitive Corner

SINGLE OPTION CORRECT TYPE:

1. **Correct answer – A) Rusting of iron**
   Explanation: Rusting occurs gradually over time and is harmful (non-desirable).
   Example: Iron gates turning reddish-brown over months.
2. **Correct answer – C) Melting of ice**
   Explanation: Non-periodic changes do not repeat regularly. Melting happens randomly depending on temperature.
3. **Correct answer – B) Burning of coal for energy**
   Explanation: Desirable changes are useful. Burning coal produces energy for cooking and electricity.
4. **Correct answer – B) Ripening of a mango**
   Explanation: Ripening occurs gradually over days → slow change.
5. **Correct answer – A) Physical change**
   Explanation: No new substance is formed; water changes from liquid → gas.
6. **Correct answer – D) Because it combined with oxygen from the air**
   Explanation: Burning magnesium is a chemical reaction. Magnesium combines with oxygen to form magnesium oxide, adding the mass of oxygen.
7. **Correct answer – B) Chemical change**
   Explanation: Rusting forms a new substance (iron oxide).
8. **Correct answer – B) Formation of gas or fizzing**
   Explanation: Gas, fizzing, color change, or heat/light are indicators of chemical changes.
9. **Correct answer – B) Cooking an egg**
   Explanation: Physical change: shape and size change.
   Chemical change: proteins denature to form new substance.
10. **Correct answer – B) Melting of ice**
    Explanation: Ice → water is a state change, no new substance formed.
11. **Correct answer – A) Physical change; aluminum can be reused**
    Explanation: Melting changes state only, not composition → recyclable.
12. **Correct answer – B) Chemical, desirable**
    Explanation: Pesticides chemically react to kill pests → beneficial chemical change.
13. **Correct answer – B) Chemical, slow**

    Explanation: Slow chemical change: oxidation causes color and chemical composition change.
14. **Correct answer – C) Physical change; fast change**

    Explanation: Water boils quickly (fast) → physical change (liquid → gas).

15. **Correct answer – C) Periodic physical change**
Explanation: Rotation repeats regularly → periodic; no new substance → physical.
16. **Correct answer – D) Desirable energy conversion**
Explanation: Sunlight is converted into usable electricity → useful physical/electrical energy change.
17. **Correct answer – A) Physical, non-desirable**
Explanation: Melting is physical and useful to eat.
18. **Correct answer – A) Fast chemical change due to ethylene gas**
Explanation: Ethylene from bananas speeds up mango ripening → fast chemical change.
19. **Correct answer – B**
Explanation: Rusting produces a new substance (iron oxide) and requires oxygen and water; cannot be reversed.
20. **Correct answer – B**
Explanation: Camphor changes state from solid → gas without forming a new substance → physical change.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

21. **Correct answer – A) Formation of gas, C) Color change**

Explanation:
- Chemical changes produce a new substance. Common indicators: gas formation, color change, heat/light, odor change.
- Melting of wax and evaporation of water are physical changes (no new substances formed).

22. **Correct answer – A) Magnetizing iron, B) Dissolving salt in water, D) Freezing of water**
Explanation:
- Physical changes: no new substances formed; shape, state, or size may change.
- Photosynthesis is a chemical change (produces glucose and oxygen).

23. **Correct answer – A) Light or sound produced, B) Substance changes color, D) Gas formation**
Explanation:
- These are signs of chemical reactions.
- Substance melting is a physical change, not a chemical change.

24. **Correct answer – A) Rusting of iron, B) Rotting of food, D) Formation of cracks in walls**
Explanation:
- Non-desirable changes are harmful or unwanted.
- Melting ice is neutral or useful ⟶ not non-desirable..

25. **Correct answer – A) Boiling water for tea, B) Ripening of fruits using ethylene gas**
Explanation:
- Fast changes occur quickly.
- Boiling water is fast and useful.
- Ethylene gas speeds up ripening ⟶ fast and beneficial.
- Melting ice is slow; rusting of iron is slow and harmful.

26. **Correct answer – A) Rusting of iron, B) Rotting of food**
Explanation:
- Slow changes take time to occur.
- Rusting and food spoilage are harmful → slow & non-desirable.
- Boiling water and growth of plants are either fast or desirable → not non-desirable.

ASSERTION AND REASON TYPE

27. **Correct answer – A) Both A and R are true and R is the correct explanation of A**

Explanation: Seasons repeat every year at fixed intervals → that is exactly what a periodic change means.

28. **Correct answer – D) A is false but R is true**

Explanation: Rusting is not desirable because it damages iron objects.

29. Correct answer – D) A is false but R is true

Explanation: Burning paper forms ash, gases, heat → new substances form → chemical change.

Reason correctly defines physical change → so A false, R true.

30. Correct answer – A) Both A and R are true and R is the correct explanation of A

Explanation: Milk converts into curd (new substance formed).

Change cannot be reversed → chemical change.

MATRIX MATCHING TYPE:

31. Correct answer

a-Q, b-R, c-P, d-S

Explanation:

A. At this point, energy is used to break the lattice structure of a solid to turn it into a liquid..

B. This is the temperature at which a gas loses enough energy for its particles to cluster into a liquid.

C. This is the temperature at which the vapor pressure of the liquid equals the surrounding pressure.

D. The temperature at which liquid particles slow down enough to form a rigid structure.

32. Correct answer

a-P, b-Q, c-S, d-R

Explanation:

A. In photosynthesis, plants convert CO_2 and water into glucose and oxygen gas.

B. Shiny grey iron reacts with oxygen and moisture to form reddish-brown iron oxide (rust).

C. Bacterial decomposition of organic matter produces foul-smelling gases (like sulfur compounds).

D. Combustion (burning) is an exothermic reaction that releases energy, often as light and heat.

COMPREHENSION TYPE

33. Correct answer – C) Melting ice

Explanation:
A physical change affects only the state or appearance of a substance without changing its chemical composition. When ice melts, it changes from solid to liquid, but it is still water (H_2O). Burning paper, rusting, and cooking involve chemical changes because new substances are formed.

34. Correct answer – B) Molecular composition

Explanation:
In a physical change, the molecules themselves do not change. For example, when ice melts, H_2O molecules remain the same. Properties like color or temperature may change temporarily, and energy may be absorbed or released, but the chemical composition stays constant.

35. Correct answer – C) They do not alter molecular composition

Explanation:
Physical changes do not produce new substances. Only the form or appearance changes, such as melting, freezing, or dissolving. They may be reversible or irreversible, and they do not necessarily involve heat release.

36. Correct answer – B) Boiling water

Explanation:
Boiling water is a physical change because the water changes from liquid to gas, but its chemical composition (H_2O) remains the same. Burning wax, rusting iron, and cooking rice involve chemical changes because they produce new substances.

37. **Correct answer – C) Photosynthesis**
Explanation:
Photosynthesis is a chemical change because plants convert carbon dioxide and water into glucose and oxygen, forming new substances. Melting, condensation, and freezing are physical changes since no new substances are produced.
38. **Correct answer – B) Chemical changes involve rearrangement of atoms**
Explanation:
In chemical changes, the atoms of the reactants rearrange to form new products with different properties. Physical changes only affect the state or appearance without changing the chemical structure. Energy may be absorbed or released in chemical changes.

Integer type:

39. **Correct answer –4 times**
Explanation: Spring, Summer, Rainy/Monsoon, Winter
40. **Correct answer –15**
Explanation: Slow changes = 2 × fast changes
Let fast **=** x
Then slow = 2x
Total = x + 2x = 3x
3x = 15; x = 5
41. **Correct answer – Number of slow changes = 3**
Explanation: Growth of a tree → slow
Rusting of iron → slow
Burning of paper → fast
Formation of curd → slow
Explosion of a cracker → fast
42. **Correct answer – Fast changes = 6**

Total changes = 18

Explanation: Slow changes are 6 more than fast changes

Let fast = x
Then slow = x + 6

Total = x + (x + 6) = 18
2x + 6 = 18
2x = 12; x = 6

Previous year questions:

1. **Correct answer – D (I, II and IV only)**

Explanation: Statements I, II, and IV represent chemical changes

2. **Correct answer – A (Neha and Sachin)**
Explanation: Neha: "The salty water makes the process of rusting faster." — Correct. Salt acts as an electrolyte, speeding up the electrochemical reaction of rusting.
Sachin: "For rusting, the presence of both oxygen and water (or water vapour) is essential." — Correct. Rusting cannot occur if either air or moisture is missing.
3. **Correct answer – B (I and III only)**
Explanation: Manure formation from leaves is a chemical change: Correct. Decomposition by microorganisms breaks down the complex organic matter in leaves into simpler substances, creating a new product (manure).
Breakdown of ozone to oxygen is a chemical change: Correct. Ozone (O3) decomposes into oxygen molecules (O2), which is a change in chemical composition.
4. **Correct answer – B (Raw egg to boiled egg)**
Explanation: Raw egg to boiled egg: Irreversible. Boiling causes the proteins in the egg to denature and solidify; you cannot turn a solid, cooked egg back into a liquid, raw egg.
5. **Correct answer – is B. 2, 4, and 5 only**

Explanation:

Physical, reversible. Crystallization is a physical process; you can re-dissolve the crystals to get the solution back.

Chemical, irreversible. Glucose reacts with oxygen to form CO_2 and water, releasing energy. You cannot simply reverse this in the body.

Physical, reversible. Sugar dissolves in water without changing its identity. You can recover the sugar by evaporating the water.

6. **Correct answer – A (I and IV only)**

   Explanation: No new substances is formed during a chemical change: Incorrect. The defining feature of a chemical change is that at least one new substance is always formed.

   Chemical changes are generally reversible: Incorrect. Chemical changes are generally irreversible because the internal structure of the substance has been altered.

7. **Correct answer – C (P-iii, Q-i, R-ii)**

   Explanation: P. Separating stones from rice: This is a Physical, reversible change (iii)

   Q. Curdling of milk: This is a Chemical, irreversible change

   R. Breaking of a flower vase: This is a Physical, irreversible change (ii)

8. **Correct answer – C (R)**

   R (Submerged hull): This portion stays under the water. Saltwater is an electrolyte that significantly accelerates the chemical reaction of rusting. Because this part is constantly in contact with salty sea water, it undergoes the most aggressive corrosion.

9. **Correct answer – C (All are physical changes)**
   - Beating aluminium to make foil: Physical change (only shape changes).
   - Cutting a log of wood: Physical change (only size changes).
   - Heating an iron nail till red hot: Physical change (only color/temperature change; no new substance formed).
   - Crystallisation of pure copper sulphate**:** Physical change (separation technique).

10. **Correct answer – A**

    Explanation: Cooking of food has the following characteristics:
    - It is a Chemical change (composition changes).
    - It is a Desirable change (we want it to happen).
    - It is a Non-periodic change (it doesn't happen at fixed regular intervals like a heartbeat).
    - It is an Irreversible change (you cannot un-cook food).

2. Fibre to Fabric

Solutions

Worksheet – 1

1. **Correct answer – A**
   The wool coat (hair) cut from a sheep's body in one piece is called a fleece.
2. **Correct answer – C**
   Shearing is the process of cutting off the sheep's wool (fleece), usually done in spring.
3. **Correct answer – A**
   Scouring removes dust, dirt, grease, and dried sweat from the fleece.
4. **Correct answer – A**
   Fibres obtained from plants (like cotton) and animals (like wool and silk) are called natural fibres.
5. **Correct answer – C**
   During scouring, fleece is washed with water and soap to remove impurities.
6. **Correct answer – D**
   Sorting separates fibres based on their texture and quality.
7. **Correct answer – C**
   Wool traps air, which acts as an insulator and keeps us warm in winter.
8. **Correct answer – D**
   Wool is mainly obtained from the hair of sheep.
9. **Correct answer – C**
   Wool is obtained from the hair (fleece) of the sheep.
10. **Correct answer – D**
    The Lohi breed is found in Rajasthan and Punjab and is known for good-quality wool.

Worksheet – 2

1. **Correct answer –D**

   Silk fibre is made of a protein called fibroin, which is secreted by the silkworm.

2. **Correct answer – B**

   The rearing of silkworms for producing silk is known as sericulture.

3. **Correct answer – A**
   The silky covering spun by the silkworm around itself is called a cocoon.
4. **Correct answer – D**
   The cocoon protects the silkworm during its pupa stage of development.
5. **Correct answer – C**
   If the cocoon is broken by the emerging moth, the continuous silk filament is cut into short pieces, lowering its quality.
6. **Correct answer – A**
   The process of unwinding silk threads from the cocoon for use as silk is called reeling.
7. **Correct answer – A**
   Mulberry silk is the most common type of silk and is obtained from silkworms that feed on mulberry leaves.
8. **Correct answer – B**
   Farmers commonly refer to silk moth eggs as seeds.
9. **Correct answer – C**
   Silk moth eggs hatch properly at an ideal temperature of about 25 °C to 27 °C.
10. **Correct answer – C**
    Silkworms mainly feed on mulberry leaves for their growth and development.

Competitive Corner

1. **Correct answer – B**
   Wool is made of keratin, a fibrous structural protein also found in human hair and nails.
2. **Correct answer – B**
   Selective breeding improves the quality and quantity of soft under hair (fine fleece), which produces better wool.
3. **Correct answer –C**

   Shearing is done in spring so sheep remain comfortable during summer and do not suffer from heat.

4. **Correct answer – B**
   Burrs are unwanted plant materials that get trapped in the fleece and are removed during processing.
5. **Correct answer – C**
   Carding uses metal teeth to separate, straighten, and untangle wool fibres.
6. **Correct answer – B**
   The natural crimp (waviness) of wool makes it elastic and easy to knit, and helps trap air for warmth.
7. **Correct answer – B**
   The harness lifts warp threads up and down to allow the shuttle to pass during weaving.
8. **Correct answer – C**
   Stifling uses steam or heat to kill the pupa so the silk thread remains unbroken.
9. **Correct answer – B**

   Sericin is the sticky protein that binds silk fibres together in the cocoon.

10. **Correct answer – C**

    About 2,000–3,000 cocoons are needed to produce 500 grams of silk because each cocoon yields a very fine thread.

11. **Correct answer – B**

    Wool is an animal fibre made of the protein keratin. It is obtained from the fleece of sheep, goats, yaks, etc., and is commonly used to make warm clothes like sweaters and coats.

12. **Correct answer – B**

    Stifling is the process of heating cocoons to kill the pupa inside. If the moth emerges naturally, it breaks the silk filament, so it does not remain long and continuous. Unwinding silk filaments from the cocoon is called reeling.

ONE OR MORE THAN ONE CORRECT OPTION TYPE:

13. **Correct answer**

    A, C, D. Wool is made of keratin, a protein found in animal hair. Wool fibres trap air, which acts as an insulator. Because trapped air reduces heat loss, woollen clothes keep the body warm.

14. **Correct answer**

    A, B, C. Scouring removes dirt and grease from fleece. Cleaning of burrs removes small twigs and plant materials. Carding separates and straightens fibres.

15. **Correct answer**

    A, B, D. Stifling kills the pupa so the moth does not break the silk filament. Reeling is the process of unwinding silk from the cocoon. Several (3–8) filaments are twisted together to make silk yarn.

16. **Correct answer**

    A, B, C. Mulberry silk, Tasar (Tussar) silk, and Muga silk are natural silk varieties.

17. **Correct answer**

    A, B, D. Skin allergies may occur due to contact with fibres or chemicals. Respiratory problems can result from inhaling fine fibres. Exposure to vapours and fine fibres can affect health.

ASSERTION AND REASON TYPE:

18. **Correct answer – A**

    Wool traps air, and since air is a poor conductor of heat, it prevents body heat from escaping, keeping us warm.

19. **Correct answer – A**

    Stifling is done before reeling because if the moth comes out naturally, it breaks the continuous silk filament.

20. **Correct answer – A**

    Shearing is done in spring because sheep need their fleece to survive the cold winter.

21. **Correct answer – D**

    Wool is obtained from the fine soft under hair of sheep, not the coarse outer hair.

22. **Correct answer – D**

    Reeling is the process of unwinding silk from cocoons, while scouring removes grease and dirt from fleece.

MATRIX MATCHING TYPE:

23. Correct answer

a-R, b-Q, c-P, d-S. Yaks are found in cold regions like Tibet and Ladakh. Angora goats are reared in Jammu and Kashmir for fine wool (mohair). Llamas are native to South America. Alpacas are also found in South America and are known for soft wool.

24. Correct answer

a-Q, b-P, c-R, d-S. Lohi sheep produce good quality wool. Rampur Bushair sheep are known for brown coloured fleece. Nali sheep produce wool mainly used for making carpets. Marwari sheep produce coarse wool suitable for rough use.

COMPREHENSION TYPE:

Passage - I

25. Correct answer – B

Shearing is done in spring so that sheep do not suffer from cold after their fleece is removed.

26. Correct answer – C

Scouring is the process of washing the sheared fleece to remove dirt, grease, and dust.

27. Correct answer – B

Carding is the process of combing wool fibres with metal teeth to separate and straighten them.

Passage – II

28. Correct answer –B

The larval stage of the silk moth is called a caterpillar or silkworm.

29. Correct answer – C

The cocoon protects the silk moth during its pupa stage.

30. Correct answer – C

Reeling is the process of extracting silk thread from the cocoon.

Previous Year Questions:

1. Correct answer – A

Silk is the strongest natural fibre, fabrics made from silk are smooth with a unique lustre, and silk is obtained by killing the silkworm (an insect) during the process of sericulture.

2. Correct answer – D

Reeling is the process of extracting silk fibre from the cocoon. Mulberry leaves are the primary food of silkworms. So both options (B) and (C) are correct.

3. Correct answer – C

Lohi sheep are found in Rajasthan and Punjab.

4. Correct answer – C

Arranging two sets of yarn to make fabric is weaving, not knitting. Mohair is obtained from the Angora goat, not the Angora rabbit (rabbit wool is called angora). Rearing silkworms for silk production is called sericulture.

5. Correct answer – A

Woollen clothes keep us warm because wool fibres trap air in the spaces between them. Air is a poor conductor of heat, so the trapped air prevents body heat from escaping, keeping us warm in cold weather.

6. Correct answer – B

Llama and alpaca are South American camelids, not breeds of camels; they yield wool but are distinct from true camels. The silkworm cocoon secretes a protein fibre (fibroin) that hardens on exposure to air, forming silk fibre. Anthrax is an occupational disease in the wool industry caused by Bacillus anthracis.

3. Wastewater Story

Solutions

Worksheet – 1

1. **Correct answer – B**
   Untreated human excreta contain harmful microorganisms that spread diseases, so they are a health hazard.
2. **Correct answer – A**
   Cooking oil and fats solidify and block pipes, so avoiding them prevents drains from getting choked.
3. **Correct answer – A**
   Sewage mainly consists of wastewater from homes and industries, so it is a liquid waste.
4. **Correct answer – C**
   Water used for washing, bathing, and flushing becomes dirty and is called wastewater.
5. **Correct answer – D**
   Both onsite sewage disposal and a sewer system are techniques used to improve sanitation.
6. **Correct answer – B**
   Rainwater itself is clean and not a direct cause of water pollution (though it may carry pollutants).
7. **Correct answer – C**
   Oils and fats should not be disposed of in drains because they cause blockages.
8. **Correct answer – C**
   Industrial activities produce large amounts of wastewater containing chemicals and pollutants.
9. **Correct answer – B**
   Groundwater is being pumped out faster than it can be naturally replenished by rainfall, causing the water table to fall.
10. **Correct answer –C**
    Biological impurities include microorganisms like bacteria that can cause diseases such as cholera and typhoid.

Worksheet – 2

1. **Correct answer – A**
   Sewage contains a complex mixture of suspended solids, organic and inorganic impurities, and disease-causing bacteria. These harmful substances that make water unfit for use are collectively known as contaminants.
2. **Correct answer – B**
   A WWTP is a facility designed to remove physical, chemical, and biological contaminants from sewage so that the water can be safely returned to the environment.
3. **Correct answer – C**
   Just like an electrical grid, the underground network of pipes that transports sewage from its source (homes/factories) to a treatment plant is called a sewerage system, and the individual pipes are sewers.
4. **Correct answer – D**
   After sludge (settled solid waste) is treated and dried, it is rich in organic matter and nutrients, making it an excellent natural fertilizer or manure for crops.
5. **Correct answer – C**
   Solid kitchen waste (like fats and food scraps) hardens and clogs pipes, while certain chemicals can kill the bacteria that help break down waste, leading to buildup and blockages.
6. **Correct answer – B**
   In this tank, the speed of incoming wastewater is decreased. This allows heavier objects like sand, grit, and pebbles to settle at the bottom by gravity—a process called sedimentation.

7. **Correct answer – B**
   Chlorine and ozone are powerful oxidizing agents. They are used in the final stages of treatment to kill harmful bacteria and microbes, ensuring the water is disinfected before release.
8. **Correct answer – C**
   Sludge is the human waste and organic matter that settles at the bottom of the large sedimentation tanks during the primary stage of treatment at a WWTP.
9. **Correct answer – C**
   Treatment involves physical (grit removal), biological (decomposition by bacteria), and chemical (chlorination) steps. Evaporation is a part of the natural water cycle, but it is not a standard industrial process used to treat large volumes of sewage.
10. **Correct answer – C**
    Cholera, Typhoid, and Hepatitis are all water-borne diseases caused by consuming or coming into contact with contaminated water. A hemorrhage refers to heavy bleeding, which is usually the result of an injury or a specific medical condition, not sewage-borne pathogens.

Competitive Corner:

SINGLE OPTION CORRECT TYPE:

1. **Correct answer – C**
   Approximately 97% of Earth's water is in the oceans and is salty, making it unfit for drinking.
2. **Correct answer – A**
   Groundwater is being pumped out faster than rainfall can replenish it, causing the water table to fall.
3. **Correct answer – C**
   Trees help the soil retain moisture; without them, the ground cannot recharge its water levels effectively.
4. **Correct answer – B**
   Grey water comes from sinks, showers, and washing machines and contains fewer pathogens than toilet waste.
5. **Correct answer – C**
   These substances, along with metals and mineral salts, are categorized as inorganic impurities in sewage.
6. **Correct answer – C**
   Yellow water refers specifically to urine that is collected through separate channels and not mixed with other waste.
7. **Correct answer – A**
   Sewerage is the entire underground network of interconnected pipes that transports waste to treatment plants.
8. **Correct answer – B**
   Manholes are placed every 50 to 60 meters to allow workers access for inspection and cleaning.
9. **Correct answer – C**
   Depending on how much it is used, a septic tank needs the solid sludge removed every one to three years
10. **Correct answer – C**
    These toilets use chemicals in a reservoir below the seat to reduce odors and carry out partial disinfection.
11. **Correct answer – B**
    While solids settle at the bottom, skimmers are used to remove substances like oil and grease that float on the surface.
12. **Correct answer –**
    B. Anaerobic bacteria decompose the sludge, producing biogas which can then be used for electricity or heating.
13. **Correct answer – C**
    Air bubbles provide oxygen to aerobic bacteria, helping them grow and consume human and food waste in the water.
14. **Correct answer – C**
    Responsible citizens should report leaks or open drains to local authorities to ensure proper sanitation and health

15. **Correct answer – B**
The VSS implemented water harvesting techniques to solve severe groundwater shortages in the Medak region.
16. **Correct answer – C**
Check dams are built across streams to slow down water, allowing it more time to seep into the ground and recharge aquifers.
17. **Correct answer – D**
Sand and grit removal physically separates heavy solids like sand, grit, and faecal particles from wastewater by allowing them to settle.
18. **Correct answer – C**
Biogas (mainly methane) is burned as a fuel, so it is used as a source of energy.
19. **Correct answer – B**
Sludge is decomposed by anaerobic bacteria in the absence of oxygen, producing biogas.
20. **Correct answer – A**
Aerobic bacteria use oxygen to break down organic matter in clarified water, helping to clean it further.
21. **Correct answer – C**
A skimmer is used to remove floatable solids like oil and grease from the surface of wastewater.
22. **Correct answer – B**
Cholera, jaundice, and typhoid are water-borne diseases caused by improper sewage disposal, but heart attack is not.
23. **Correct answer – B**
Ozone is used in water treatment to kill microorganisms. It is not essential for breathing, does not make up 3% of air, though it does absorb ultraviolet rays.
24. **Correct answer –C**
The by-products of wastewater treatment include sludge and biogas.

ONE OR MORE THE ONE CORRECT OPTION TYPE:

25. **Correct answer**

A, C, D. Organic impurities include pesticides, herbicides, urea, and food wastes. Phosphates and nitrates are specifically listed as inorganic impurities.

26. **Correct answer**

A, B, C, D. Manholes are vertical holes used for inspection and cleaning. They are strategically placed at junctions or where the sewer line changes direction.

27. **Correct answer**

A, C, D. Paints, solvents, and motor oils kill microbes that digest waste. Fats and oils can clog soil pores, reducing their filtering effectiveness. While soft toys/cotton shouldn't be thrown in drains because they choke them, they aren't specifically cited as killers of helpful microbes.

28. **Correct answer**

A, C, D. The VSS used percolation tanks, contour trenches, and rock fill dams to improve groundwater levels. Distillation is not mentioned as a method used by the community.

29. **Correct answer**

A, B, D. In the first sedimentation tank, solids settle as sludge which is removed by a scraper, and floating grease is removed by a skimmer. Chlorination is a chemical process that happens much later in the treatment cycle.

ASSERTION AND REASON TYPE:

30. **Correct answer – A**
The United Nations General Assembly proclaimed this decade specifically to focus efforts on improving global access to safe water.
31. **Correct answer – A**
The biological process relies on oxygen to increase bacterial activity so the microbes can break down harmful matter remaining in the water.
32. **Correct answer – B**

Both statements are true. Untreated waste causes pollution and disease, and most Earth water is indeed saline. However, the salinity of ocean water does not explain why human waste is a health hazard.

33. Correct answer –D

The assertion is false; oils should never be thrown down the drain as they cause blockages. The reason correctly describes the negative impact of oils on drainage systems.

34. Correct answer – C

The assertion is true as septic tanks are onsite disposal systems for areas without sewerage. The reason is false because septic tanks are actually low-cost and are specifically used where there is *no* connection to a sewer line.

MATRIX MATCHING TYPE:

35. Correct answer

a-R, b-T, c-Q, d-P, e-S. Black water is defined as highly contaminated liquid waste originating specifically from toilet fixtures. Contour trenches are dug along the slopes of hills specifically to slow down rainwater and prevent the soil from washing away. Grey water refers to wastewater from non-toilet sources; it is generally cleaner and carries fewer disease-causing germs (pathogens). Check dams are built across moving streams to slow down the water flow, which helps the water seep into the ground (infiltration). Rock fill dams are used in areas prone to gullies (deep channels cut by running water) to trap sediment and manage water flow.

36. Correct answer

a-P, b-R, c-S, d-T, e-Q. Organic impurities consist of carbon-based waste products, including human and animal excreta, oil, and food remains. Inorganic impurities are mineral-based substances such as nitrates, phosphates, and heavy metals. Phosphorus and nitrogen are specifically listed as the primary nutrients found in sewage that can encourage plant/algal growth. Bacteria in sewage are the primary cause of severe waterborne diseases like cholera and typhoid. Other microbes, specifically protozoa, are responsible for causing waterborne diseases like amoebic dysentery.

COMPREHENSION TYPE

Passage - I

37. Correct answer – B

22 March is celebrated as World Water Day to bring awareness to the importance of safe drinking water.

38. Correct answer – C

The efforts during the International Decade for Action aimed to reduce the number of people without access to safe drinking water by half.

39. Correct answer – D

Wastewater is defined as the dirty water from sinks, showers, and laundries; glaciers are a source of fresh water, even if inaccessible

Passage – II

40. Correct answer – B

Disposing of chemicals like medicines and solvents in drains is discouraged because they kill microbes responsible for breaking down organic waste.

41. Correct answer – C

Sewage is defined as liquid waste that includes wastewater from homes and rainwater that has run down the street.

42. Correct answer – C

Choked drains prevent the free flow of oxygen, which hampers the natural degradation process of waste.

Integer Type:

43. 1 unit. According to the table, groundwater, rivers, and lakes together make up 1% of Earth's total water. If total water is 100 units, then 1% of 100 = 1 unit.

44. 95%.

Ocean water = 97%
Glaciers and polar ice = 2%
Difference = 97 – 2 = 95%.

4. Acids and Bases

Solutions

Worksheet – 1

1. **Correct answer – B**

   Indicators show different colours in acidic and basic solutions to help identify them.

2. **Correct answer – C**

   Litmus solution is purple (mauve) in a neutral solution.

3. **Correct answer – C**

   Spinach contains oxalic acid naturally.

4. **Correct answer – C**

   Acidity means the number of replaceable H^+ ions. Since $Al(OH)_3$ is a base and has 3 OH^- groups, its basicity is 3.

5. **Correct answer – B**

   China rose (hibiscus) indicator turns dark pink in acidic solutions.

6. **Correct answer – A**

   Phenolphthalein is a synthetic (man-made) indicator prepared in laboratories.

7. **Correct answer – D**

   Lemon contains citric acid, which gives it a sour taste.

8. **Correct answer – B**

   Calcium hydroxide decomposes to Cao and H_2O on heating.

9. **Correct answer – B**

   Litmus is a natural dye obtained from lichens (a combination of algae and fungi).

10. **Correct answer**

    Non-metal oxides react with water to form acids. Carbon dioxide reacts with water to form carbonic acid.

Worksheet – 2

1. **Correct answer – A**

   Acidic salts produce acidic solutions in water and turn blue litmus red.

2. **Correct answer – B**

   The first part of a salt's name comes from the base (metal or ammonium ion). For example, in sodium chloride, sodium comes from sodium hydroxide.

3. **Correct answer – A**

   $NaOH + HCl \rightarrow NaCl + H_2O$. The salt formed is sodium chloride (NaCl).

4. **Correct answer – A**

   A hydrated salt contains water of crystallization. $CuSO_4 \cdot 5H_2O$ has five water molecules attached.

5. **Correct answer – A**

   Anhydrous salts do not contain water of crystallization.

6. **Correct answer – A**

   The Taj Mahal is made of marble (calcium carbonate), which reacts with acids present in acid rain and gets damaged.

7. **Correct answer – C**

   Acidic waste can harm aquatic organisms. Neutralizing the acid makes the water safer for plants and animals.

8. **Correct answer – C**

   This reaction is called neutralization. Acid + Base → Salt + Water.

9. **Correct answer – A**

   The second part of a salt's name comes from the acid. For example, in sodium chloride, chloride comes from hydrochloric acid.

10. **Correct answer – A**

    Sour foods contain acids that can react with metals like brass, forming harmful substances.

Competitive Corner:

SINGLE OPTION CORRECT TYPE:

1. **Correct answer – A**

   Litmus turns red in acidic solutions.

2. **Correct answer – C**

   China rose (hibiscus) indicator turns green in basic solutions.

3. **Correct answer – A**

   Vinegar contains acetic acid, which gives it a sour taste.

4. **Correct answer – C**

   Bases feel soapy or slippery when touched.

5. **Correct answer – A**

   Neutralisation is an exothermic reaction that releases heat.

6. **Correct answer – C**

   Neutral substances do not show acidic or basic properties.

7. **Correct answer – A**

   Acids generally taste sour.

8. **Correct answer – A**

   Caustic soda is the common name for sodium hydroxide.

9. **Correct answer – C**

   Ant stings contain formic acid. Baking soda is a base and helps neutralize the acid, reducing irritation and burning.

10. **Correct answer – C**

    Quick lime (calcium oxide) is basic in nature and neutralizes acidic soil, making it suitable for plant growth.

11. **Correct answer – D**

    Window cleaners commonly contain ammonium hydroxide (NH_4OH), which helps remove grease and dirt.

12. **Correct answer – A**

    A base that dissolves in water to produce OH^- ions is called an alkali.

13. **Correct answer – B**

    Turmeric turns red in a basic solution, so substance X must be a base. Baking soda is a base, therefore X can be baking soda. Phenolphthalein remains colourless in an acidic solution, so substance Y must be an acid. Hydrochloric acid is an acid, so Y can be hydrochloric acid. Hence, substances X and Y are baking soda and hydrochloric acid respectively.

14. **Correct answer – B**

    according to the Arrhenius definition, a base produces hydroxyl ions (OH^-) as the only negatively charged ions in aqueous solution. Statement 2 is also true since bases like sodium hydroxide and ammonium hydroxide feel soapy or slippery to touch. However, the slippery nature does not explain why bases produce OH^- ions in water.

Multiple

15. **Correct answer**

    A, C, D. Acids react with active metals (like zinc, magnesium) to produce hydrogen gas and a salt. Acids react with metal carbonates to produce salt, water, and carbon dioxide gas.

16. **Correct answer**

    A, B, C. Calcium hydroxide, sodium bicarbonate and magnesium hydroxide are bases.

17. **Correct answer**

    A, C, D. Litmus changes to red in acidic solutions. Methyl orange Changes to pink, orange, or red in acidic solutions. Hibsicus changes to magenta in acidic solutions.

18. **Correct answer**

    A, C. Acidic salts are formed by partial neutralization of a polybasic acid and still contain replaceable hydrogen (H^+) ions. NaH_2PO_4 contains hydrogen and is formed by partial neutralization of phosphoric acid (H_3PO_4). Na_2HPO_4 also formed by partial neutralization and contains hydrogen.

ASSERTION AND REASON TYPE:

19. Correct answer – A

When salts dissolve in water, they form free-moving ions (cations and anions). These ions carry electric charge, which allows the solution to conduct electricity.

20. Correct answer – B

Sour foods should not be stored in copper or brass because acids react with these metals and form harmful substances. Although vessels are coated with tin (kalai), this is a preventive measure and not the reason why sour foods should not be stored in them.

21. Correct answer – C

Neutralization releases heat, so the container feels warm. So, neutralization is an exothermic reaction. The reason is false because it incorrectly states that heat is absorbed.

22. Correct answer – A

NaOH completely ionises in water to give Na^+ and OH^- ions, producing a high concentration of OH^- ions, which makes it a strong base.

23. Correct answer – D

HCl does not contain oxygen, so it is a hydracid, not an oxyacid. The reason correctly states that oxyacids contain oxygen atoms.

MATRIX MATCHING TYPE:

24. Correct answer

a-S, b-R, c-P, d-Q. Nitric acid has the chemical formula HNO_3. Sodium hydroxide is represented by the formula NaOH. Sulphuric acid has the formula H_2SO_4. Ammonium hydroxide is written as NH_4OH.

25. Correct answer

a-S, b-Q, c-P, d-R. Jaggery has a sweet taste. Yogurt tastes sour due to lactic acid. Coffee has a bitter taste. Soy sauce tastes salty because it contains salt.

COMPREHENSION TYPE:

Passage – I

26. Correct answer – C

Turmeric is a natural indicator that turns red in a basic (alkaline) solution.

27. Correct answer – D

Turmeric does not change colour in an acidic solution and remains yellow.

28. Correct answer – B

Bases generally have a bitter taste, unlike acids which taste sour.

Passage – II

29. Correct answer – D

Ant stings contain formic acid, which causes irritation and a burning sensation.

30. Correct answer – B

Citrus fruits contain citric acid, which gives them their sour taste.

31. Correct answer – C

Lactic acid is formed during the fermentation of milk, giving curd and sour milk their sour taste.

Integer Type:

32. Correct answer – 3

Basicity is the number of replaceable H^+ ions in an acid. H_3PO_4 has three ionisable hydrogen atoms, so its basicity is 3.

33. Correct answer – 5

Pentahydrate means five water molecules are attached. Its formula is $CuSO_4 \cdot 5H_2O$.

34. Correct answer – 1

Acidity of a base is the number of OH^- ions it can furnish. NaOH gives one OH^- ion, so its value is 1.

Previous Year Questions:

1. **Correct answer – A**

   Reaction between dilute sodium hydroxide (NaOH) and dilute sulphuric acid (H_2SO_4) is given below.

   $$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

   This is a neutralisation reaction and it is exothermic (releases heat).

2. **Correct answer – B**

   Phenolphthalein is colourless in acidic medium (turns pink in basic medium). Vitamin C is ascorbic acid. Sulphuric acid is present in lead-acid batteries. Tamarind contains tartaric acid.

3. **Correct answer – D**

   Phenolphthalein ⟶ Pink in base

   China rose ⟶ Green in base

   Methyl orange ⟶ Red in acid

   Lime water is base. Milk of magnesia is base and curd is acid.

4. **Correct answer – C**

   I. Amla juice ⟶ Acidic ⟶ Red

   II. Detergent solution ⟶ Basic ⟶ Yellow

   III. Soda water ⟶ Slightly acidic (carbonic acid present) ⟶ Red

   IV. NaCl solution ⟶ Neutral ⟶ No change

5. **Correct answer – B**

   Ammonium chloride (strong acid + weak base) ⟶ Acidic salt

   Sodium acetate (weak acid + strong base) ⟶ Basic salt

   Lemon contains acid, which cleans tarnished copper by reacting with copper oxide.

6. **Correct answer – B**

   a. Soft drink → Acidic
   b. Tamarind juice → Acidic
   c. Milk of magnesia → Basic
   d. Tomato juice → Acidic
   e. Lime water → Basic
   f. NaCl solution → Neutral
   g. Window cleaner → Basic
   h. Calamine solution → Basic

   Turmeric paper turns red in basic solution. Base turns red litmus blue while acid turns blue litmus red. Methyl orange turns red in acidic solution and yellow in basic/neutral solutions.

7. **Correct answer – A**

   China rose indicator: Green colour in base, Magenta colour in acid.

   Calamine solution is base and lemon juice is acid.

8. **Correct answer – C**

   Acids produce hydrogen ions (H^+) in aqueous solution. Acids are sour in taste; bases are bitter.

9. **Correct answer – B**

   Soap solution is basic, so it turns methyl orange yellow. Curd is acidic, so it turns China rose indicator magenta.

10. **Correct answer – B**

    When magnesium burns, it forms magnesium oxide, which dissolves in water to form magnesium hydroxide (a base). So, it turns red litmus blue and phenolphthalein pink

5. Metals and Non-Metals

Solutions

Worksheet – 1

1. **Correct answer – B) Metalloids**
   Explanation: Metalloids, also called semimetals, exhibit properties of both metals and non-metals. For example, silicon is brittle like a non-metal but conducts electricity like a metal.
2. **Correct answer – C) Iron**
   Explanation**:** Density depends on mass per unit volume. Approximate densities:
   - Sodium ≈ 0.97 g/cm^3
   - Potassium ≈ 0.86 g/cm^3
   - Iron ≈ 7.87 g/cm^3
   - Helium ≈ 0.000178 g/cm^3
3. **Correct answer – A) Ductility**
   Explanation: Ductility is the ability of a metal to be stretched into wires without breaking. Malleability is the ability to be hammered into sheets, sonority is the ability to produce sound, and lustre is the shiny appearance
4. **Correct answer – C) Beryllium**
   Explanation**:** Beryllium is a metal with a very strong metallic bond, giving it a high melting point (~1287 °C) and boiling point (~2470 °C). Sodium, potassium, and mercury have much lower melting/boiling points.
5. **Correct answer – B) Sodium and Potassium**
   Explanation**:** Most metals have high melting points, but alkali metals like sodium and potassium are soft and melt at low temperatures (Sodium: 98 °C, Potassium: 63 °C).
6. **Correct answer – C) A homogeneous mixture of a metal with other metals or non-metals**
   Explanation: An alloy combines metals or metals with non-metals to improve properties (e.g., steel = iron + carbon). It is not a pure metal or a compound.
7. **Correct answer – D) Helium.**
   Helium is a noble gas and is found in the free (elemental) state in nature because it is chemically inert and does not readily form compounds with other element
8. **Correct answer – B) Tensile strength**
   Explanation**:** Tensile strength is the maximum stress a material can withstand while being stretched or pulled. Metals are strong and durable, making them ideal for structural applications.
9. **Correct answer – B) Graphite**
   Explanation: Graphite has a shiny metallic-like appearance (lustre) due to its layered structure. Chlorine, oxygen, and phosphorus do not show natural lustre.
10. **Correct answer – B)Oxides and sulphides**
    Explanation**:** Metals are usually chemically reactive, so they are found as ores (oxides, sulphides) rather than free elements. Only a few metals like gold, platinum occur naturally in elemental form.

Worksheet-2

1. **Correct Answer: (B) Remove impurities from molten iron**
   Explanation: Oxygen is blown into molten iron to oxidize impurities like carbon, silicon, and phosphorus. This helps convert molten iron into steel by reducing unwanted elements.
2. **Correct Answer: (B) Prevent spoilage by providing an inert atmosphere**
   Explanation**:** Nitrogen is inert and displaces oxygen in food packages, preventing oxidation and growth of bacteria, which keeps food fresh longer.

3. **Correct Answer: (B) Sulphuric acid**

   Explanation: Sulphuric acid is called the "king of chemicals" because of its wide industrial use in fertilizers, batteries, chemical manufacturing, and petroleum refining.

4. **Correct Answer: (B) Fertilizers for root and seed development**

   Explanation: Phosphorus is essential for plant growth, especially for strong roots and seeds, and is a key ingredient in phosphate fertilizers.

5. **Correct Answer: (A) Drinking water to kill germs**

   Explanation**:** Chlorine is a disinfectant used to eliminate harmful microorganisms in water, making it safe to drink.

6. **Correct Answer: (C) Prevent iodine deficiency and goitre**

   Explanation**:** Iodine prevents thyroid gland enlargement (goitre) and deficiency-related health issues, which is why table salt is iodized.

7. **Correct Answer: (A) Teflon (non-stick coating)**

   Explanation**:** Fluorine is used to produce compounds like Teflon (PTFE) for non-stick coatings and other fluoropolymers.

8. **Correct Answer: (B) Transporting oxygen in blood via hemoglobin**

   Explanation**:** Iron is a key component of hemoglobin, which binds and carries oxygen from the lungs to tissues.

9. **Correct Answer: (B) It is lightweight and conducts electricity efficiently**

   Explanation**:** Aluminum is less dense than copper and conducts electricity well, making it ideal for overhead transmission lines where weight is a concern.

10. **Correct Answer: (B) Rechargeable batteries for EVs and electronics**

    Explanation**:** Lithium-ion batteries are widely used in electric vehicles, smartphones, laptops, and other electronics due to high energy density.

Worksheet-3

1. **Correct Answer: (C) Silicon**

   Explanation: Silicon is a semiconductor widely used in electronics. It forms the basis of computer chips, transistors, and solar cells due to its ability to control electrical conductivity.

2. **Correct Answer: (A) A semiconductor and flame retardant**

   Explanation: Antimony is used in alloys to improve hardness and in plastics as a flame retardant. It also has semiconductor applications.

3. **Correct Answer: (B) Welding and steelmaking**

   Explanation: Argon is an inert gas that creates a protective atmosphere, preventing metals from reacting with oxygen or nitrogen during high-temperature processes like welding or steelmaking.

4. **Correct Answer: (B) Cancer radiotherapy and radon spas**

   Explanation**:** Radon is radioactive. It has been used medically in radiotherapy and in certain spas for therapeutic purposes, though its use is now limited due to health risks.

5. **Correct Answer: (B) Insulating ability and light-emitting properties**

   Explanation: Krypton is a noble gas used in high-performance lighting, lasers, and double-glazed windows for insulation due to its low thermal conductivity and inertness.

6. **Correct Answer: (B) Gradual deterioration of metals due to reactions with air, moisture, or chemicals**

   Explanation: Corrosion, such as rusting of iron, is a chemical process where metals degrade due to environmental factors.

7. **Correct Answer: (B) Oxidation of fats and oils**

   Explanation**:** Fats and oils oxidize over time when exposed to air, producing unpleasant odors and tastes, which is called rancidity.

8. **Correct Answer: (B) Prevent rancidity**

   Explanation**:** BHA (Butylated Hydroxyanisole) and BHT (Butylated Hydroxytoluene) are added to foods to slow oxidation of fats and oils, preventing rancidity.

9. **Correct Answer: (B) Cooling superconducting magnets and balloons**

   Explanation: Helium is inert and has very low boiling point. It cools superconducting magnets (like MRI machines) and inflates balloons because it is lighter than air.

10. **Correct Answer: (B) Boron**

    Explanation**:** Boron is a key component of borosilicate glass and fiberglass, providing thermal stability and strength.

Competitive corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (B) Storing food in airtight containers**

   Explanation: Rancidity occurs when fats and oils oxidize. Limiting contact with air by using airtight containers slows down oxidation and prevents rancidity.

2. **Correct Answer: (C) Xenon**

   Explanation: Xenon is heavy, inert, and can be ionized to generate thrust in ion propulsion systems used in spacecraft.

3. **Correct Answer: (B) Fiber-optic communication and infrared technology**

   Explanation: Germanium is a semiconductor that is transparent to infrared light and is used in optical fibers, infrared optics, and electronics.

4. **Correct Answer: (B) Galvanization**

   Explanation: Galvanization involves coating iron with a layer of zinc, which protects it from corrosion, especially in harsh environments like seawater.

5. **Correct Answer: (A) Radon**

   Explanation: Radon is naturally radioactive and seeps from underground. Measuring its movement helps trace underground contamination and air flow patterns.

6. **Correct Answer: (B) It produces extremely high temperatures with fuels like acetylene**

   Explanation: Oxygen supports combustion and, when combined with fuels such as acetylene, produces very high temperatures necessary for welding metals.

7. **Correct Answer: (B) Rapid freezing and preservation of biological samples**

   Explanation: Liquid nitrogen is extremely cold (-196°C), making it ideal for cryogenic freezing of food, biological samples, and medical storage.

8. **Correct Answer: (A) High electrical conductivity and corrosion resistance**

   Explanation: Copper conducts electricity very efficiently and resists corrosion, making it ideal for electrical wiring and plumbing.

9. **Correct Answer: (A) Aerospace (jet engines, airframes) and medical implants**

   Explanation: Titanium is strong, lightweight, and corrosion-resistant, which makes it ideal for aerospace components and medical implants.

10. **Correct Answer: (C) Concrete or tungsten-based shields**

    Explanation: While lead is effective, it is toxic. Tungsten and concrete provide radiation shielding without toxicity.

11. **Correct Answer: (B) Oxygen – oxidizer, Nitrogen – food preservation, Aluminum – structural metal**

    Explanation: Oxygen acts as an oxidizer in rocket fuel, nitrogen preserves food by creating an inert atmosphere, and aluminum is lightweight and strong for spacecraft structures.

12. **Correct Answer: (B) Bromine**

    Explanation: Bromine is a red-brown liquid at room temperature (~25°C). Carbon, nitrogen, and sulphur are solid or gaseous under normal conditions.

13. Correct Answer: (B) Ability to produce a ringing sound

Explanation: Sonority refers to the property of metals to produce a clear ringing sound when struck.

14. Correct Answer: (B) Sodium

Explanation: Alkali metals like sodium and potassium are very soft and can be cut with a knife, unlike iron, copper, or gold.

15. Correct Answer: (C) Silver

Explanation: Silver has the highest electrical conductivity of all metals, even higher than copper or aluminum. Mercury is a liquid at room temperature and conducts poorly.

ONE OR MORE THAN ONE CORRECT TYPE

16. Correct Answer: (B) Argon, (C) Krypton, (D) Xenon

Explanation**:**

- Argon is used in incandescent and fluorescent bulbs to prevent filament oxidation.
- Krypton is used in high-performance lighting and photography flash lamps.
- Xenon is used in high-intensity lamps, headlights, and lasers.
- Helium is not typically used in lighting.

17. Correct Answer: (A) Adding antioxidants like BHA or BHT, (B) Storing food in airtight containers, (D) Using nitrogen gas to prevent oxygen exposure

Explanation:

- Rancidity occurs due to oxidation of fats and oils.
- Antioxidants slow down oxidation.
- Airtight containers prevent air (oxygen) contact.
- Nitrogen displaces oxygen, creating an inert atmosphere to prevent spoilage.
- Applying grease to metals is unrelated to food preservation.

18. Correct Answer: (B) Immune function and wound healing

Explanation:

- Zinc is an essential trace element. It supports the immune system, aids in wound healing, and is required for enzymatic reactions.
- Oxygen transport is carried out by iron in hemoglobin. Muscle contraction mainly involves calcium, sodium, and potassium. Digestion involves enzymes, not specifically zinc.

19. Correct Answer: (C)Food preservation

Explanation:

Krypton is chemically inert and extremely expensive to extract from the atmosphere. Using it for food preservation would be like using a designer watch as a paperweight—it's overkill and way too pricey compared to nitrogen or carbon dioxide.

20. Correct Answer: (A) Vulcanization of rubber, (B) Fertilizer production, (D) Fireworks and matches

Explanation:

- Vulcanization of rubber**:** Sulphur is used to cross-link rubber molecules, improving elasticity and strength.
- Fertilizer production: Sulphur compounds are important in fertilizers like ammonium sulphate.
- Fireworks and matches**:** Sulphur acts as a fuel and combustion enhancer.
- Electrical wiring**:** Sulphur is not used here; metals like copper/aluminum are used.

ASSERTION AND REASON TYPE

21. Correct Answer: (D) Assertion is incorrect but Reason is correct

Explanation:

- Assertion is wrong: Oxygen is not used to preserve food; in fact, oxygen promotes spoilage because it allows oxidation and microbial growth.
- Reason is correct: Oxygen is indeed essential for respiration in living organisms.
- They are not connected in a cause-effect way.

22. Correct Answer: (A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

Explanation:

- Safety matches contain phosphorus compounds (usually red or white phosphorus).
- Reason correctly explains the assertion: Phosphorus ignites easily on friction and produces light/heat, which is why it is used in matches.

23. Correct Answer: (D) Assertion is incorrect but Reason is correct

Explanation:

- Aluminum is indeed lighter, but titanium is preferred for jet engine parts because it is stronger, withstands higher temperatures, and resists corrosion.
- So the assertion is incorrect, but the reason is true.

24. Correct Answer: (A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

Explanation:

- Krypton is used in double-glazed windows because it is a heavy, inert gas that reduces heat transfer.
- Low thermal conductivity directly explains its high insulating ability.

25. Correct Answer: (A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

Explanation:

- Metals can be hammered into sheets (malleable) or drawn into wires (ductile).
- Reason explains it perfectly: Metallic bonds are non-directional, allowing layers of metal ions to slide without breaking the overall bond.

MATRIX MATCHING TYPE

26. Correct Answer:

1→C, 2→B, 3→D, 4→A

Explanation:

1. Nitrogen is inert and displaces oxygen to prevent oxidation and spoilage in packaged foods.
2. Aluminum is light, strong, and corrosion-resistant, making it ideal for aircraft structures.
3. Lithium is used in lithium-ion batteries for EVs and electronics.
4. Iron (often as steel) is strong and durable, suitable for constructing bridges and heavy structures.

27. Correct Answer:

1→B, 2→A, 3→C, 4→D

Explanation:

1. Oxygen is essential for breathing and combustion processes.
2. Nitrogen prevents oxidation and microbial growth in packaged food.
3. Sulfur is used to cross-link rubber molecules, improving elasticity and strength.
4. Chlorine kills harmful microorganisms, making water safe to drink.

COMPREHENSION TYPE

28. Correct Answer: (B) Aluminum

Explanation**:** Aluminum has low density (lightweight) and forms a protective oxide layer that prevents corrosion. These properties make it ideal for aircraft bodies.

29. Correct Answer: (B) Conducts electricity efficiently

Explanation**:** Copper has very high electrical conductivity and low resistance, making it perfect for electrical wiring. Cost and non-toxicity are not the main reasons.

30. Correct Answer: (A) It helps in respiration as part of hemoglobin

Explanation: Iron is a key component of hemoglobin in blood, which transports oxygen from lungs to body tissues.

31. Correct Answer: (B) Rusting of iron

Explanation: Corrosion specifically refers to the deterioration of metals. Rusting is the corrosion of iron due to oxygen and moisture.

32. Correct Answer: (B) Adding antioxidants

Explanation: Antioxidants slow down oxidation of fats and oils, preventing unpleasant smell and taste in food.

33. Correct Answer: (C) Galvanization with zinc

Explanation: Galvanization coats iron with zinc. Zinc prevents contact with air and moisture and protects iron from rusting.

INTEGER TYPE

34. Correct Answer: 5

Explanation**:** The noble gases naturally present in Earth's atmosphere are helium, neon, argon, krypton, and xenon. (Generally not present in the atmosphere due to its rapid decay.)

35. Correct Answer: 1

Explanation: Only mercury is liquid at room temperature (~25 °C). Other metals like gallium and cesium melt near room temperature but are solid at 25 °C.

36. Correct Answer: 11

Explanation: Gaseous non-metals at room temperature include:
Hydrogen, nitrogen, oxygen, fluorine, chlorine, and the noble gases helium, neon, argon, krypton, xenon, radon → total 11.

37. Correct Answer: 0.002%

Explanation**:** Nitrogen is abundant in the atmosphere (~78%) but is present only in trace amounts in the Earth's crust by mass.

Previous year questions

1. Correct Answer:D. Carbon, magnesium only

Carbon:
Marked: × malleable, ✓ ductile, × lustrous, ✓ hard
Incorrect — carbon (a non-metal) is not ductile. Only diamond (a form of carbon) is hard, but carbon is not generally ductile or malleable.

Magnesium:
Marked: × malleable, × ductile, × lustrous, ✓ hard
Incorrect — magnesium is a metal and is lustrous, malleable, and ductile.

2. Correct Answer: D. a only

a) Metal
K, Zn, C, Fe, Cu, P
Correct classification → metals vs non-metals properly shown.

6. Basic Concepts of Chemistry

Solutions

Worksheet – 1

1. **Correct Answer: B**
   1 mole of any substance contains Avogadro's number of particles, i.e., 6.022×10^{23}atoms.
2. **Correct Answer: C**
   Atomic weight expressed in grams is called gram atomic weight, which is also known as gram atom. Both terms mean the same.
3. **Correct Answer: A**
   Molar mass of CO_2 = 44 g.
   So, 44 g CO_2 = 1 mole CO_2.
   Each CO_2 molecule contains 2 oxygen atoms → therefore, 1 mole CO_2 contains 2 moles of oxygen atoms.
4. **Correct Answer: D**
   At NTP, 1 mole gas occupies 22.4 L = 22400 mL.
   Molecules in 1 mL = $\frac{6.022\times10^{23}}{22400} \approx 2.67 \times 10^{19}$
5. **Correct Answer: C**
   Molar mass of $CaCO_3$ = 100 g.
   50 g $CaCO_3$ = 0.5 mole.
   Each $CaCO_3$ has 3 oxygen atoms.
   So, oxygen atoms = $0.5 \times 3 = 1.5$moles of O atoms.
   Number of atoms = $1.5 \times 6.022 \times 10^{23} = 9.035 \times 10^{23}$.
6. **Correct Answer: C**
   Given: Molecular weight = V.D × 2
   $= 24 \times 2 = 48$
7. **Correct Answer: B**
   At STP, 1 mole of any gas occupies 22.414 L (approximately 22.4 L).
8. **Correct Answer: B**
   Hydrogen exists as H_2 molecule.
   Atomic mass of H = 1.008
   Molar mass of $H_2 = 1.008 \times 2 = 2.016$g/mol.
9. **Correct Answer: D**
   Chlorine exists as Cl_2.
   Atomic mass of Cl ≈ 35.5
   Molar mass of $Cl_2 = 35.5 \times 2 = 71$g.
10. **Correct Answer: C**
    Avogadro number is defined as the number of atoms present in 12 g (0.012 kg) of carbon-

Worksheet – 2

1. **Correct Answer: A**
   $CaO + H_2O \rightarrow Ca(OH)_2$
   Two substances combine to form a single product, so it is a combination reaction.
2. **Correct Answer: A**
   When two or more substances react to form only one product, it is called a combination reaction.
3. **Correct Answer: B**
   $AgNO_3 + NaCl \rightarrow AgCl$ (white precipitate) $+ NaNO_3$
   Silver chloride (AgCl) forms a white precipitate.
4. **Correct Answer: C**
   $(A^+B^-) + (C^+D^-) \rightarrow (A^+D^-) + (C^+B^-)$
   Exchange of ions takes place between two compounds, so it is a double displacement reaction.
5. **Correct Answer: D**
   Among Pt, Cu, Zn, and Mg, magnesium (Mg) is the most active metal according to the reactivity series.
6. **Correct Answer: C**
   Reactions that proceed by absorption of light energy are called photochemical reactions.
7. **Correct Answer: B**
   $CuO + H_2 \rightarrow Cu + H_2O$
   CuO is reduced to Cu, so it acts as the oxidizing agent (it provides oxygen).
8. **Correct Answer: B**
   A chemical reaction is characterized by the formation of new substance(s) with different properties.

9. **Correct Answer: C**
   Electrolysis of water requires the passage of electric current to decompose water into hydrogen and oxygen.
10. **Correct Answer: B**
    In a chemical equation, reactants are written on the left-hand side and products on the right-hand side.

Worksheet – 3

1. **Correct Answer: B**
   Nitrogen forms different oxides (e.g., NO, NO_2, N_2O_5) where oxygen combines with nitrogen in different simple whole-number ratios. This follows the law of multiple proportions.
2. **Correct Answer: D**
   First oxide:
   O = 0.5 g, X = 0.5 g
   Second oxide:
   O = 1.6 g, X = 4.0 − 1.6 = 2.4 g
   O per gram of X:
   First = 0.5 / 0.5 = 1
   Second = 1.6 / 2.4 = 2/3
   Ratio = 1 : 2/3 = 3 : 2 (simple whole number ratio)
3. **Correct Answer: D**
   First compound: O% = 100 − 42.9 = 57.1
   Second compound:
   O% = 100 − 27.3 = 72.7
   O per 1 g C:
   First = 57.1 / 42.9 ≈ 1.33
   Second = 72.7 / 27.3 ≈ 2.67
   Ratio = 1.33 : 2.67 ≈ 1 : 2
4. **Correct Answer: D**
   Tin combines with chlorine in two different ratios (2 Cl and 4 Cl atoms), showing different simple whole-number proportions.
5. **Correct Answer: D**
   Molecular mass calculation gives 28 parts nitrogen and 80 parts oxygen.
6. **Correct Answer: C**
   Different substances are made of different types of atoms and molecules, which give them different physical and chemical properties.
7. **Correct Answer: A**
   Mass of reactants:
   1.7 g + 0.585 g = 2.285 g
   Mass of products:
   1.435 g + 0.85 g = 2.285 g
8. **Correct Answer: A**
   Chemical equations are balanced because matter is neither created nor destroyed during a reaction.
9. **Correct Answer: B**
   Hydrogen and oxygen form water (H_2O) and hydrogen peroxide (H_2O_2), showing multiple proportions.
10. **Correct Answer: C**
    John Dalton proposed the law of multiple proportions as part of his atomic theory.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A**
   The atomic mass of one oxygen atom (O) is 16 u as per the periodic table.
2. **Correct Answer: A**
   One mole of hydrogen atoms has a mass equal to its atomic mass in grams, i.e., 1.008 g.
3. **Correct Answer: A**
   One mole of any substance contains Avogadro's number (6.02×10^{23}) of particles.
4. **Correct Answer: D**
   Molar mass of N_2 = 28 g/mol
   Molar mass of O_2 = 32 g/mol
   Number of moles ∝ 1 / molar mass
   So ratio = 32 : 28 = 8 : 7.
5. **Correct Answer: A**
   1 mole of water (H_2O) contains:
   - 1 mole of water molecules → which has 2 moles of H atoms + 1 mole of O atoms = 3 moles of atoms → (ii) correct.
   - It contains 1 mole of oxygen atoms, not 16 g of oxygen separately → (iv) incorrect in this context.
   - Total electrons in 1 H_2O molecule = 10, so 1 mole contains 1 mole of electrons, not 10 moles → (iii) incorrect.

   Thus, (i) and (ii) are correct.

6. Correct Answer: B

Molar mass of CO = 28 g/mol
7 g CO = 7/28 = 0.25 mol
28 g CO = 1 mol

At STP, volume ∝ number of moles.
So ratio = 0.25 : 1 = 1 : 4.

7. Correct Answer: C

Number of molecules ∝ number of moles.

- 28 g CO → Molar mass = 28 → 1 mole
- 46 g C_2H_5OH → Molar mass = 46 → 1 mole
- 36 g H_2O → Molar mass = 18 → 2 moles
- 54 g N_2O_5 → Molar mass = 108 → 0.5 mole

Since 36 g H_2O has 2 moles, it has the largest number of molecules.

8. Correct Answer: D

1 mole of any gas at STP occupies 22.4 L.
2 moles × 22.4 L = 44.8 L.

9. Correct Answer: A

At STP, 22.4 L = 1 mole.
5.6 L = 5.6 / 22.4 = 0.25 mole.
Molar mass = 7.5 g / 0.25 = 30 g/mol.
Molar mass of NO = 30 g/mol, so the gas is NO.

10. Correct Answer: B

From the reactivity series:
Al is more reactive than Zn, Zn more than Fe, and Au is least reactive.
Thus decreasing order: Al > Zn > Fe > Au.

11. Correct Answer: D

$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$

- Ions exchange places → Double displacement reaction.
- Insoluble AgCl forms as a solid → Precipitation reaction.

12. Correct Answer:B

H_2SO_4 (acid) reacts with NaOH (base) to form salt (Na_2SO_4) and water → Neutralization reaction.

Also, ions exchange between acid and base → Double displacement reaction.

It is neither decomposition nor addition reaction.

13. Correct Answer: C

LHS: 2 H, 2 O
RHS: 2 H, 1 O
Oxygen atoms are not equal, so it is unbalanced.

14. Correct Answer: B

MnO_2 speeds up the decomposition of H_2O_2 into water and oxygen without being consumed.

15. Correct Answer: B

one element replaces another in a compound (displacement reaction).

16. Correct Answer: C

The arrow (→) in a chemical equation means "yields" or "forms." It shows that the reactants on the left produce the products on the right.

17. Correct Answer: B

By the law of conservation of mass:
Mass of Mg + Mass of O = Mass of MgO
12 g + x = 20 g
x = 20 − 12 = 8 g

18. Correct Answer: A

Total mass of products = 8.8 g + 5.4 g = 14.2 g
Mass of hydrocarbon = 3 g, so oxygen mass = 14.2 − 3 = 11.2 g.
The total mass of reactants equals the total mass of products, illustrating the law of conservation of mass.

19. Correct Answer: A

Pure carbon dioxide always contains carbon and oxygen in a fixed mass ratio (27.27% C and 72.73% O), irrespective of its source. This supports the law of constant (definite) proportion.

20. Correct Answer: B

The law of constant proportion was given by Joseph Proust.

21. Correct Answer: A

In CO_2, atomic mass of carbon = 12 and oxygen = 16 × 2 = 32.

Thus, the mass ratio of carbon to oxygen is 12 : 32.

ONE OR MORE THAN ONE CORRECT OPTION TYPE:

22. Correct Answer: B, C, D.

Absolute mass of one Na atom is 23 atomic mass units (since atomic mass of Na = 23 u).

Absolute mass of one atom = atomic mass × 1 amu in grams.

1 amu = 1.66×10^{-24}g.

1 amu = 1.66×10^{-27}kg.

23. Correct Answer: A, C, D

Helium is monoatomic.

1 mole of He = 1 mole of atoms = Avogadro number of atoms.

4 L = 1 mole, so 11.2 L = 0.5 mole of H_2 molecules.

Each H_2 has 2 atoms ⟶ 0.5 × 2 = 1 mole of atoms.

Molar mass of CH_4 = 16 g

3.2 g = 0.2 mole of molecules

Each molecule has 5 atoms → 0.2 × 5 = 1 mole of atoms.

24. Correct Answer: A, B, D.

1 mole contains Avogadro number of molecules. Molar mass of NH_3 = 14 + 3 = 17 g.

1 mole of any gas at STP occupies 22.4 L (22400 mL).

25. Correct Answer: A, B, D

Sodium reacts vigorously with cold water to form NaOH and H_2. Calcium reacts with cold water to form $Ca(OH)_2$ and H_2. Zinc can also displace hydrogen from water.

26. Correct Answer: A, B

One compound breaks into simpler substances on heating. Oxidation states of Mg, C, and O remain unchanged.

27. Correct Answer: A, B, C

One compound breaks into simpler substances ⟶ thermal decomposition. Single compound splits on heating ⟶ decomposition. Water breaks into simpler substances using electricity ⟶ electrolytic decomposition.

28. Correct Answer: A, B, C

Law of conservation of energy is not a law of chemical combination.

29. Correct Answer: A, D

Same elements (N and O) combine in different ratios. N_2O_4 is a dimer of NO_2 (same ratio), so does not illustrate law.

ASSERTION AND REASON TYPE:

30. Correct Answer: A

Both statements are true. The mole is defined based on the number of atoms in 12 g of carbon-12, which equals Avogadro number. Hence, the reason correctly explains the assertion.

31. Correct Answer: C

CO and CO_2 illustrate the law of multiple proportions because carbon combines with oxygen in two different ratios (1:1 and 1:2). However, CO and CO_2 do not have the same molecular mass (CO = 28 u, CO_2 = 44 u). Therefore, the reason is false.

32. Correct Answer: A

The reaction $2Na + Cl_2 \rightarrow 2NaCl$ is a combination reaction because two different substances (sodium and chlorine) combine to form a single product (sodium chloride). The reason correctly explains why it is a combination reaction.

33. Correct Answer: D

A chemical equation is balanced according to the law of conservation of mass, not the law of definite proportion. The law of definite proportion states that a compound always contains the same elements in a fixed ratio by mass, which is true. Hence, the assertion is false but the reason is true.

34. Correct Answer: B

Platinum (Pt) does not displace hydrogen from dilute acids because it is less reactive than hydrogen and lies below hydrogen in the reactivity series.

The reason correctly defines a displacement reaction, but it does not explain why platinum cannot displace hydrogen. Hence, both are true, but the reason is not the correct explanation.

MATRIX MATCHING TYPE:

35. Correct Answer:

a-R, b-S, c-Q, d-P.

Molar mass of a gas = 2 × vapour density. Avogadro's number is 6.023×10^{23} particles per mole. At NTP, 1 mole of any gas occupies 22.4 L = 22400 cm^3. Gram atomic mass of carbon equals its atomic mass in grams, i.e., 12 g.

36. Correct Answer:

a-R, b-S, c-P, d-Q. Two or more substances combine to form a single product. One compound breaks down into simpler substances. A more reactive element (Na) displaces hydrogen from water. A more reactive element (Na) displaces hydrogen from water.

COMPREHENSION TYPE:

Passage – I

37. Correct Answer: D

3.011×10^{23} atoms of hydrogen = $\frac{3.011}{6.022} =$ 0.5mole of H atoms.

Since hydrogen gas is H_2, 2 atoms = 1 molecule, so moles of $H_2 = \frac{0.5}{2} = 0.25$mole.

At NTP, 1 mole of gas occupies 22.4 L.

Volume = $0.25 \times 22.4 = 5.6$ L.

38. Correct Answer: B

Molar mass of sulphur (S) = 32 g/mol. Therefore, mass of 1 mole of sulphur atoms = 32 g.

39. Correct Answer: A

One mole of any substance contains Avogadro's number (6.02×10^{23}) of particles. Hence, one mole of sodium represents 6.02×10^{23} sodium atoms.

Passage – II

40. Correct Answer: D

According to the law of conservation of mass, mass of reactants must be equal to mass of products. Here, 36 g of reactants gives only 32 g of products, which means 4 g is missing. Hence, it violates the law of conservation of mass.

41. Correct Answer: B

The law of conservation of mass states that total mass of reactants equals total mass of products.

Here, 6 g + 16 g = 22 g, which exactly equals the mass of CO_2 formed. So this is the best example.

42. Correct Answer: C

Mass of reactants = 5.3 g + 6 g = 11.3 g

Mass of products = 8.2 g + 2.2 g + 0.9 g = 11.3 g

Since mass is conserved, the correct total mass is 11.3 g

INTEGER TYPE:

43. Correct Answer: 4 g

One mole contains Avogadro number (6.022×10^{23}) of atoms.

The molar mass of helium = 4 g/mol, so 1 mole (Avogadro number) weighs 4 g.

44. Correct Answer: 4 g

Vapour density (VD) = Molar mass / 2.

Molar mass of SO_3 = 80 g/mol.

VD = 80/2 = 40.

Given VD = 10X ⇒ 10X = 40 ⇒ X = 4 g.

45. Correct Answer: 4 g

6×10^{24} atoms ≈ 10 moles (since Avogadro number = 6×10^{23}).

10 moles weigh 200 g ⇒ 1 mole (atomic mass) = 20 g.

For diatomic gas, molar mass = 2 × 20 = 40 g.

Given 10X = 40 ⇒ X = 4 g.

46. Correct Answer: 3

Zn, Na, and K are above hydrogen in the reactivity series and can displace hydrogen from acids. Cu, Ag, and Pt cannot.

47. Correct Answer: 2

(ii) $K + NaOH \rightarrow KOH + Na$ ✓ (single displacement)

(iii) $Fe + CuSO_4 \rightarrow Cu + FeSO_4$ ✓ (single displacement)

Others are combination, double displacement, or decomposition reactions.

48. Correct Answer: 2

In SO_2, mass of S = 32, mass of O_2 = 32 (16 ×

2).

Ratio S : O = 32 : 32 = 1 : 1.

Sum of ratios = 1 + 1 = 2